

Challenge Problems Week 4

Superdense Coding

UCQ Education Team

Quantum teleportation lets us transmit one qubit by sending two classical bits. The “opposite” of this is *superdense coding*, which lets us transmit two classical bits by sending one qubit. Imagine that Alice and Ryan each have one qubit in an entangled pair $\frac{1}{\sqrt{2}}(|0_A 0_R\rangle + |1_A 1_R\rangle)$.

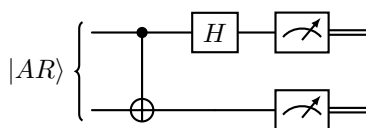
Problem 1. Using *only operations on her own qubit*, show how Alice can manipulate the pair into any one of the four states

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|0_A 0_R\rangle \pm |1_A 1_R\rangle) \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|0_A 1_R\rangle \pm |1_A 0_R\rangle)$$

Now imagine Alice sends her qubit to Ryan.

Problem 2 (finally one for the engineers!) Which qubit architecture should Alice use to send the qubit? (It’s the one most commonly used for quantum networking.)

Ryan then performs the following set of operations (known as a *Bell basis measurement*).



Problem 3. What is the measurement outcome for each of the four states in Problem 1? (4 outcomes mean 2 classical bits!)